

WNN from Cell Ranger ARC reanalyze on RNA-Harmony and ATAC-Harmony

Leiden r2 knn=30

EDA for percentage of neurons, habenula and thalamus

Dataset: RNA log-normalized with harmony; and ATAC-LSI with Harmony

Reference: Integrated (Data-Driven and Literature-Based) gene markers reference (469 genes with no repeat)

text in green Have Hb mixed with other cell-types

- Sample cluster < 10 cells
- Total cells by cluster < 1000
- Percentage of cell-types < 2%
- Clusters with ONLY one cell-type
- Hb gene-markers domains the cluster

Top 50 DEG (Wilcox test)

seurat_clusters	S03 Hb_r	S04 Hb_r	S05 Hb_r	S06 Hb_r	S07 Hb_r	S08 Hb_r	S09 Hb_r	S10 Hb_r	S11 Hb_r	S12 Hb_r	total_clust.x	Perc.Cluster	cell_types
1	966	597	296	574	515	245	219	255	137	102	3906	7.01	DD_Inhib.Thal (10), DD_LHb (1), LB_excitatory_neuron (1), LB_inhibitory_neuron (2), LB_Thalamus/MDm (1)
2	295	67	90	145	895	373	843	240	323	100	3371	6.05	DD_Oligo (5), LB_neuron (2), LB_oligodendrocyte (1)
3	255	6	368	446	125	13	1902	143	10	3	3271	5.87	DD_Astrocyte (1), DD_Endo (4), DD_Excit.Thal (1)
4	9	2048	72	54	144	295	7	46	82	154	2911	5.23	DD_Excit.Thal (6), DD_Inhib.Thal (1), DD_LHb (2), DD_OPC (1), LB_LHB neuron specific (1)
Hb_5	59	6	2	149	267	573	19	779	596	321	2771	4.98	DD_Excit.Thal (1), DD_LHb (10), LB_Hb neuron specific (2), LB_LHB neuron specific (1)
6	3	1009	116	398	229	324	62	221	171	134	2667	4.79	DD_Excit.Thal (6), DD_OPC (1)
Hb_7	144	2	3	7	798	319	14	418	286	619	2610	4.69	DD_MHb (10), DD_OPC (1), LB_Hb neuron specific (1)
8	21	105	16	134	272	209	116	364	918	382	2537	4.56	DD_Astrocyte (1), DD_Excit.Thal (3), DD_Inhib.Thal (1), DD_LHb (7), DD_OPC (1)
9	67	587	57	91	549	381	70	139	123	366	2430	4.36	DD_Excit.Thal (4), DD_OPC (4)
Hb_10	78	0	1	4	290	300	11	532	799	329	2344	4.21	DD_Excit.Thal (1), DD_LHb (1), DD_MHb (7), DD_OPC (1), LB_Hb neuron specific (2)
Hb_11	47	28	17	13	700	162	30	151	583	481	2212	3.97	DD_Excit.Thal (1), DD_MHb (17), LB_Hb neuron specific (1)
12	249	16	77	1654	95	0	81	4	0	11	2187	3.93	DD_Excit.Thal (5), DD_Inhib.Thal (1), DD_LHb (3), DD_OPC (1)
13	0	1249	96	206	60	213	20	91	52	49	2036	3.66	NA
Hb_14	32	40	12	13	193	182	8	550	744	252	2026	3.64	DD_Excit.Thal (1), DD_MHb (9), LB_MHb neuron specific (1)
15	70	7	96	610	125	7	545	138	24	3	1625	2.92	DD_Endo (2), DD_Excit.Thal (8), LB_Thalamus broad (1)
Hb_16	15	0	0	3	175	361	4	323	650	76	1607	2.89	DD_Excit.Thal (1), DD_LHb (2), DD_MHb (8), LB_Hb neuron specific (1)
17	66	78	89	319	232	20	653	52	8	70	1587	2.85	DD_Excit.Thal (11), LB_excitatory_neuron (1), LB_Hb neuron specific (1), LB_Thalamus broad (1)
Hb_18	27	9	1	92	76	185	11	300	714	120	1535	2.76	DD_Endo (1), DD_Excit.Thal (1), DD_Inhib.Thal (1), DD_LHb (11), LB_LHB neuron specific (2)
19	1	39	4	1313	5	1	10	1	3	3	1380	2.48	DD_Inhib.Thal (13), LB_inhibitory_neuron (2), LB_Thalamus/MDm (1)
20	483	90	26	33	189	70	70	170	172	40	1343	2.41	DD_Astrocyte (14), LB_astrocyte (2), LB_neuron (1)
21	101	6	239	21	142	119	40	112	395	166	1341	2.41	DD_Astrocyte (3), DD_Excit.Thal (1), LB_astrocyte (1)
22	705	15	24	31	228	79	121	52	48	24	1327	2.38	DD_Oligo (2), LB_neuron (2), LB_oligodendrocyte (1)
Hb_23	20	19	30	176	71	134	2	341	421	55	1269	2.28	DD_LHb (7), LB_LHB neuron specific (2)
Hb_24	4	3	0	1	166	29	1	27	581	13	825	1.48	DD_LHb (8), DD_MHb (4)
25	10	192	129	261	5	13	2	76	8	11	707	1.27	DD_Excit.Thal (1), DD_OPC (1)
26	128	22	8	14	198	21	212	11	16	8	638	1.15	DD_OPC (15), LB_neuron (1), LB_oligodendrocyte_precursor (2)
27	68	11	37	6	265	3	128	32	22	15	587	1.05	DD_Microglia (14), LB_microglia (2)
28	5	3	6	12	12	4	494	2	4	1	543	0.98	DD_Inhib.Thal (15), LB_inhibitory_neuron (2), LB_neuron (1), LB_Thalamus/MDm (1)
29	47	34	60	49	43	18	39	34	12	7	343	0.62	DD_Endo (12), LB_Endo/CP (2)
Hb_30	0	0	0	2	2	2	2	200	5	0	213	0.38	DD_LHb (11), LB_Hb neuron specific (1), LB_LHB neuron specific (2)
31	208	0	0	0	0	0	1	0	0	0	209	0.38	DD_Excit.Thal (6), LB_Endo/CP (1)
32	0	1	7	184	1	0	2	0	1	0	196	0.35	DD_Endo (1), DD_Excit.Thal (3), DD_LHb (2), DD_MHb (1), DD_OPC (1)
Hb_33	4	0	0	7	17	31	0	58	53	16	186	0.33	*** DD_LHb (6)
34	8	0	0	2	68	20	3	18	62	3	184	0.33	DD_Oligo (17), LB_oligodendrocyte (1)
35	162	0	0	1	0	0	1	0	1	0	165	0.30	DD_Astrocyte (1), DD_Endo (1), DD_Excit.Thal (6), DD_OPC (2), LB_Endo/CP (1)
Hb_36	10	1	1	7	21	22	6	25	26	26	145	0.26	DD_Endo (1), DD_Excit.Thal (1), DD_MHb (6)
37	6	0	2	23	4	0	71	5	0	0	111	0.20	DD_Astrocyte (1), DD_Endo (3), DD_Inhib.Thal (2), LB_inhibitory_neuron (1)
38	0	0	0	1	0	0	104	0	0	0	105	0.19	DD_Inhib.Thal (15), LB_inhibitory_neuron (2), LB_Thalamus/MDm (1)
39	0	74	4	2	0	7	1	1	0	1	90	0.16	DD_Inhib.Thal (11), LB_inhibitory_neuron (2), LB_Thalamus/MDm (1)
Hb_40	0	0	0	0	0	0	0	0	0	84	84	0.15	*** DD_LHb (1)
41	2	0	4	1	30	0	34	1	4	0	76	0.14	DD_Microglia (10), DD_Oligo (7), LB_microglia (2), LB_oligodendrocyte (1)
42	0	0	0	0	0	0	0	2	0	0	2	0.00	NA
Total.Cells.Sample	4375	6364	1990	7059	7207	4735	5959	5914	8054	4045	NA	NA	NA
Total.Cells.ALL	4375	6364	1990	7059	7207	4735	5959	5914	8054	4045	NA	NA	NA
Percentage.Sample	100	100	100	100	100	100	100	100	100	100	NA	NA	NA

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Hb_18	27	9	1	92	76	185	11	300	714	120	1535	2.76	DD_Endo (1), DD_Excit.Thal (1), DD_Inhib.Thal (1), DD_LHb (11), LB_LHB neuron specific (2)
Hb_23	20	19	30	176	71	134	2	341	421	55	1269	2.28	DD_LHb (7), LB_LHB neuron specific (2)
Hb_24	4	3	0	1	166	29	1	27	581	13	825	1.48	DD_LHb (8), DD_MHb (4)
Hb_30	0	0	0	2	2	2	2	200	5	0	213	0.38	DD_LHb (11), LB_Hb neuron specific (1), LB_LHB neuron specific (2)
Hb_33	4	0	0	7	17	31	0	58	53	16	186	0.33	*** DD_LHb (6)
Hb_36	10	1	1	7	21	22	6	25	26	26	145	0.26	DD_Endo (1), DD_Excit.Thal (1), DD_MHb (6)
Hb_40	0	0	0	0	0	0	0	0	0	84	84	0.15	*** DD_LHb (1)

